

# Customer Shopping Behavior Report of Findings

Observations from Exploratory Data Analysis (3,900 records, 18 variables)

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Tools: Python (Pandas, Matplotlib, Seaborn) | Method: Univariate, Bivariate & Multivariate EDA

# Findings by Chart

## 1. Age Distribution

The majority of customers belong to the middle-age group, with fewer customers at the younger and older ends. The distribution is fairly balanced overall, showing the business attracts a wide age range, led by middle-aged customers.

## 2. Gender Distribution

The dataset contains significantly more male customers (~2,650) than female customers (~1,250) — males form the clear majority of the customer base.

## 3. Category Distribution

Purchase frequency differs clearly across product categories, revealing the most and least purchased categories.

## 4. Purchase Amount Distribution

Purchase amounts are spread fairly evenly across the range, with no extreme skew and no single dominant spending tier.

## 5. Purchase Amount by Gender

Median purchase amount and spread are similar across genders — spending behavior is not strongly differentiated by gender.

## 6. Average Purchase Amount by Category

Average spend varies by category, with some categories commanding a noticeably higher average purchase amount than others.

## 7. Season-wise Purchases

Purchase counts are relatively balanced across seasons, though one or two seasons show slightly higher activity, hinting at mild seasonality.

## 8. Payment Method

Certain payment methods (mainly card-based options) are used far more frequently than others, showing a clear customer preference.

## 9. Subscription Status

A noticeably smaller proportion of customers are subscribed compared to non-subscribers — subscription adoption is currently low.

## 10. Shipping Type

One or two shipping types dominate customer choice, while others are used far less often.

## 11. Review Rating Distribution

Ratings are concentrated in the mid-to-upper range, suggesting most customers are moderately to highly satisfied, with relatively few very low ratings.

## 12. Discount Applied

A substantial share of purchases are made with a discount applied, indicating price sensitivity plays a real role in purchase decisions.

### **13. Promo Code Usage**

A significant portion of customers use promo codes, closely mirroring the discount-usage pattern — promotions are a strong purchase driver.

### **14. Correlation Heatmap**

Most numerical variables show very weak or no correlation with each other. Age vs. Previous Purchases ( $\sim 0.04$ ) and Purchase Amount vs. Review Rating ( $\sim 0.03$ ) are both negligible. No strong positive or negative correlations exist among the numerical variables.

### **15. Pair Plot**

Trends, distributions, and possible outliers are visible across Age, Purchase Amount, Review Rating, and Previous Purchases, but no strong linear relationships stand out — consistent with the heatmap results.

## Summary of Findings

The dataset of 3,900 customer records was clean (no duplicates, only 0.95% missing values in Review Rating), and analysis across demographics, spending, and transaction behavior surfaced the following overall patterns:

- Customer base skews male (~68%) and middle-aged, with fairly even purchase-amount spending regardless of gender.
- Product category and average spend both vary meaningfully — a clear subset of categories drives most purchase volume and revenue.
- Seasonality exists but is mild; no single season dominates purchasing.
- Payment method and shipping type usage are concentrated in a few preferred options rather than spread evenly.
- Subscription adoption is low relative to the overall customer base — a clear untapped retention opportunity.
- Discounts and promo codes are used in a large share of transactions, showing strong price sensitivity and promotion-driven buying.
- Review ratings skew positive-to-moderate, indicating general satisfaction, though a smaller low-rating segment merits monitoring.
- Numerical variables (Age, Purchase Amount, Review Rating, Previous Purchases) show negligible correlation with one another, meaning spending and satisfaction are not simply explained by age or purchase history — segmentation should rely more on categorical behavior (category, payment, subscription, discount usage) than numeric relationships.

**Overall takeaway:** Purchasing behavior is broadly consistent across demographics, but clear differentiation exists at the category, payment, shipping, and promotion level. The biggest immediate opportunities are boosting subscription adoption and using discounts/promo codes more strategically to protect margins, while continuing to prioritize inventory and marketing around top-performing categories and preferred payment/shipping channels.